
THANUSHA A

Full-Stack (MERN) | ML-Integrated Systems

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PROFILE SUMMARY

Result-driven **Software Development Engineer** with strong expertise in **MERN stack development** and hands-on experience in **machine learning–powered web applications**. Proven ability to design scalable systems, integrate APIs, and build data-driven features with real-world impact. Passionate about product-based development, problem solving, and continuous improvement aligned with industry standards.

SKILLS

Programming: Java, Python, JavaScript, SQL, C

Frontend: React.js, HTML5, CSS3, Leaflet.js

Backend: Node.js, Express.js

Databases: MongoDB, PostgreSQL

AI / ML: Machine Learning, NLP (Transformers – basics), Model Evaluation

Tools & Platforms: Git, GitHub, Postman, Figma

Core CS: Data Structures & Algorithms, OOP, REST APIs

Soft Skills: Clear Communication, Team Collaboration, Leadership

PROJECTS

Trip Booking Fintech Platform (MERN Stack) – Ongoing (2026)

Tech Stack: MongoDB, Express.js, React.js, Node.js, JWT

 GitHub: [thanusha-stack/Trip_Booking_Mern_Website](https://github.com/thanusha-stack/Trip_Booking_Mern_Website)

- Developing a full-stack fintech trip booking and payment-based web application using the MERN stack.
- Implementing user authentication, booking management, and building core workflows for trip listings, advance booking, and transaction handling.
- Designing MongoDB schemas for users, bookings, and payment records.
- Integrating frontend and backend components for seamless data flow and state management.

Freelancer Recruitment Platform (2024)

Tech Stack: React.js, Node.js, MongoDB

 GitHub: <https://github.com/thanusha-stack/Freelancer-Website>

- Developed a full-stack platform connecting freelancers and clients for project-based hiring.
- Implemented **role-based authentication** and secure user workflows.
- Designed scalable CRUD operations for job postings and applications.
- Improved usability through structured UI flows and modular components.

Machine Learning–Based Water Potability Prediction with Geospatial Insights (2025)

Tech Stack: Python, Flask, Random Forest, Gradient Boosting, SMOTE, Leaflet.js, CPCB Dataset

 GitHub: https://github.com/thanusha-stack/water_quality_prediction

- Built an ML system to predict water safety using physicochemical parameters (pH, hardness, turbidity).
- Achieved **high prediction reliability** using ensemble models and class-imbalance handling (SMOTE).
- Integrated **interactive geospatial visualization** to display region-wise water quality.
- Enabled real-world awareness and data-driven decision-making for public health.

Map-Based News Retrieval System (2025)

Tech Stack: React.js, Leaflet.js, Node.js, PostgreSQL, NewsData.io API

 GitHub: https://github.com/thanusha-stack/Map_based_News_Retrieval_System

- Built a location-aware news platform delivering **real-time regional news**.
- Visualized news geographically using interactive maps.
- Designed backend services to efficiently store and query location-based data.
- Enhanced user engagement through personalized, region-specific content.

EDUCATION

Bachelor of Engineering(Computer Science and Engineering) SATHYAMANGALAM, BANNARI AMMAN INSTITUTE OF TECHNOLOGY 2023-2027 ● 8.08CGPA (UPTO 4th SEM).

12th Grade at Sowdambikaa Matriculation Girls Higher Secondary School ,Trichy 2022-2023 ● Prevailed an aggregate of 91.33%

11th Grade at Sowdambikaa Matriculation Girls Higher Secondary School ,Trichy 2021-2022 ● Prevailed an aggregate of 90.83%

10th Grade at Fatima Matriculation Higher Secondary School ,Cuddalore 2020-2021